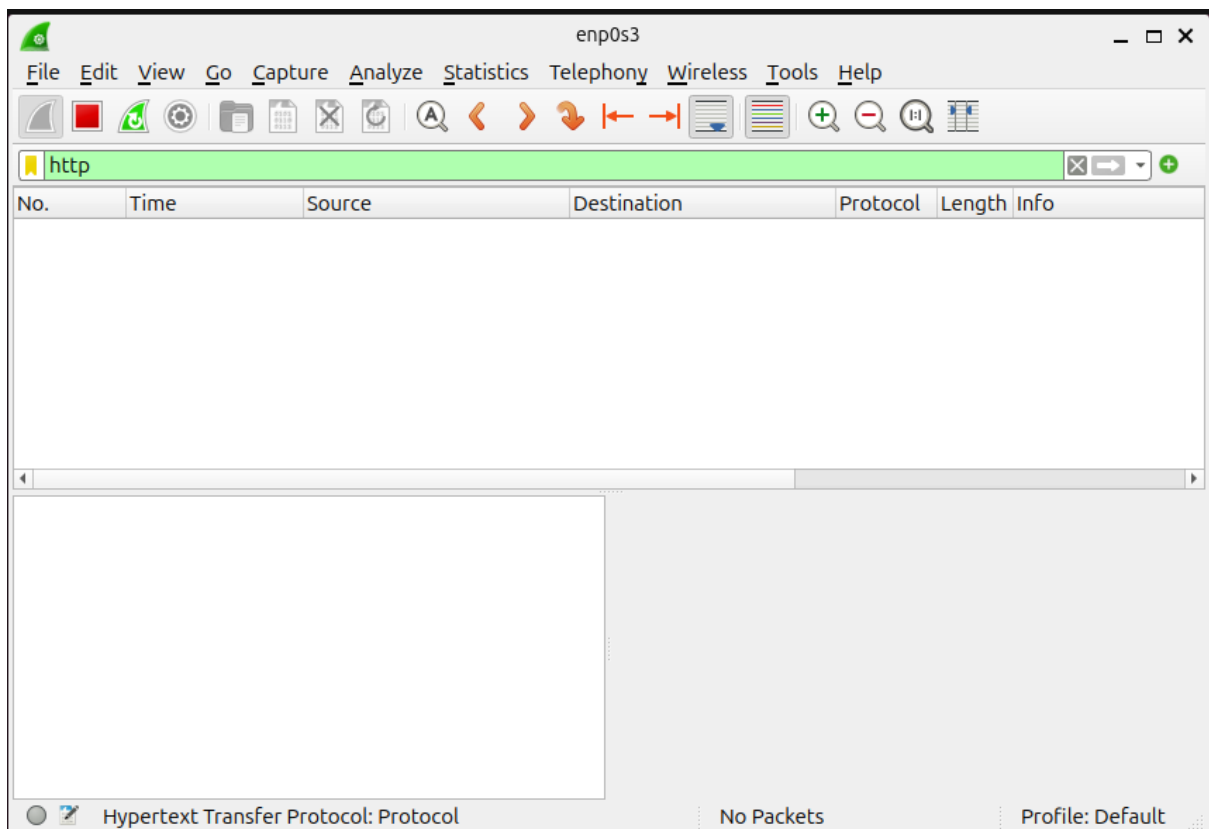
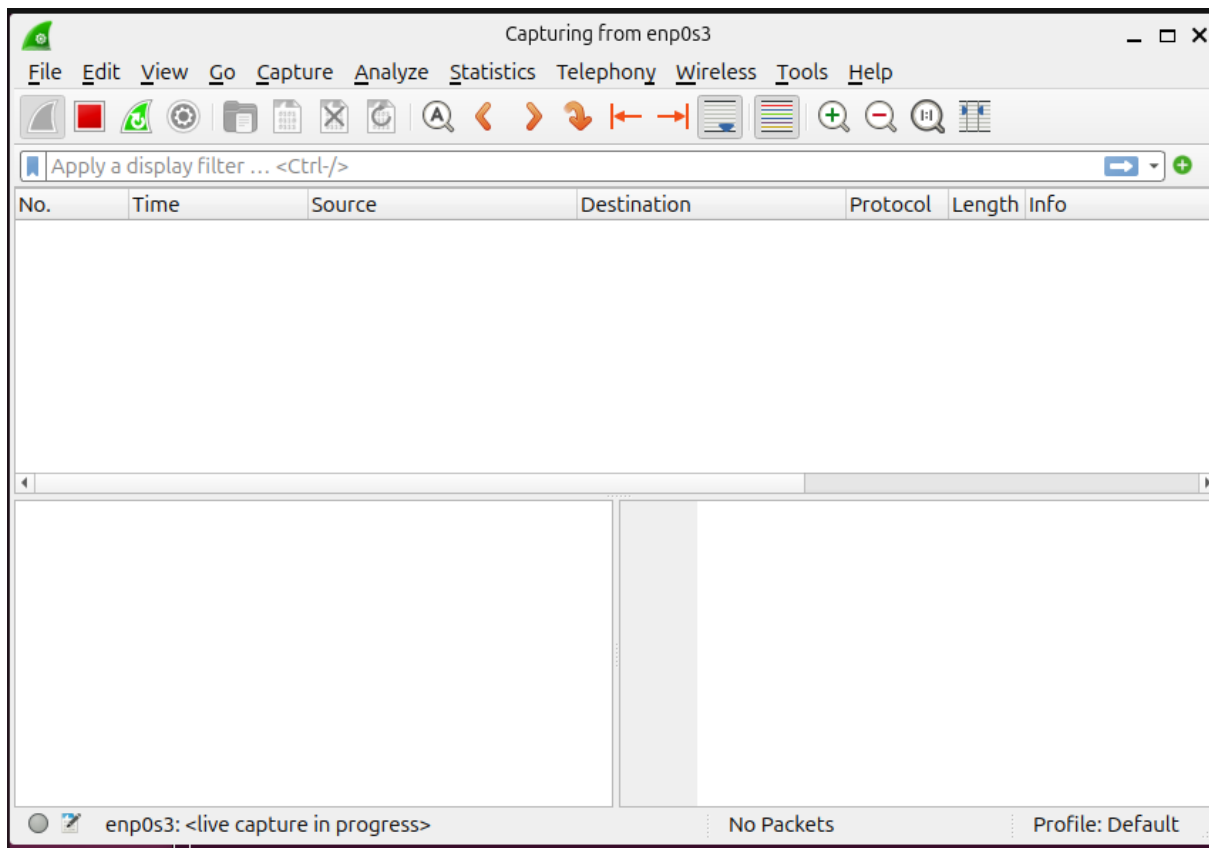
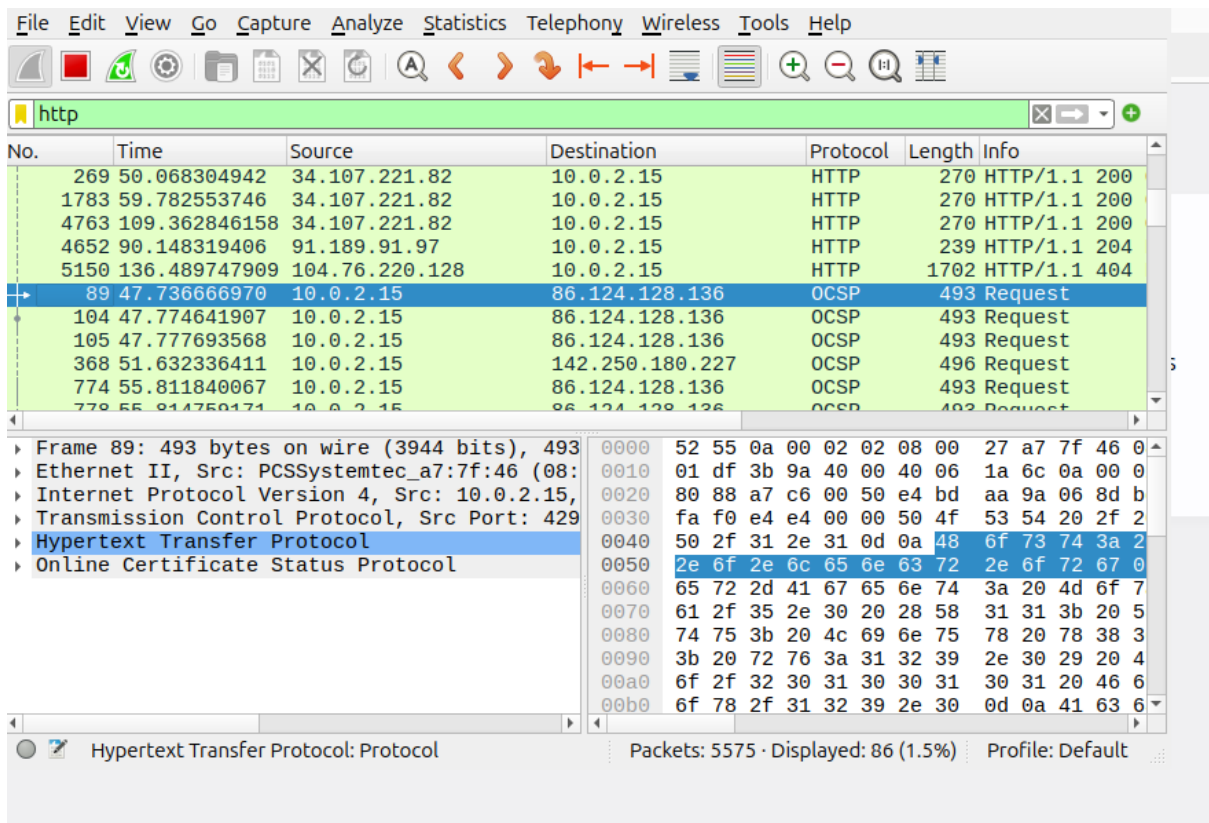
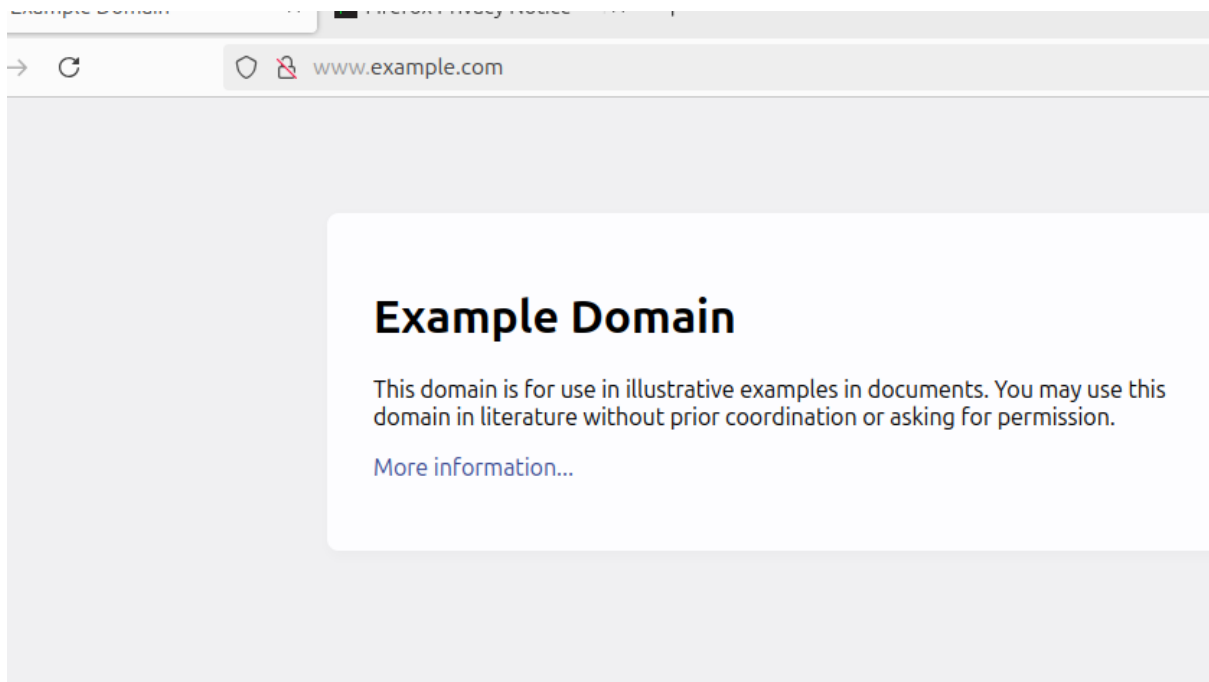
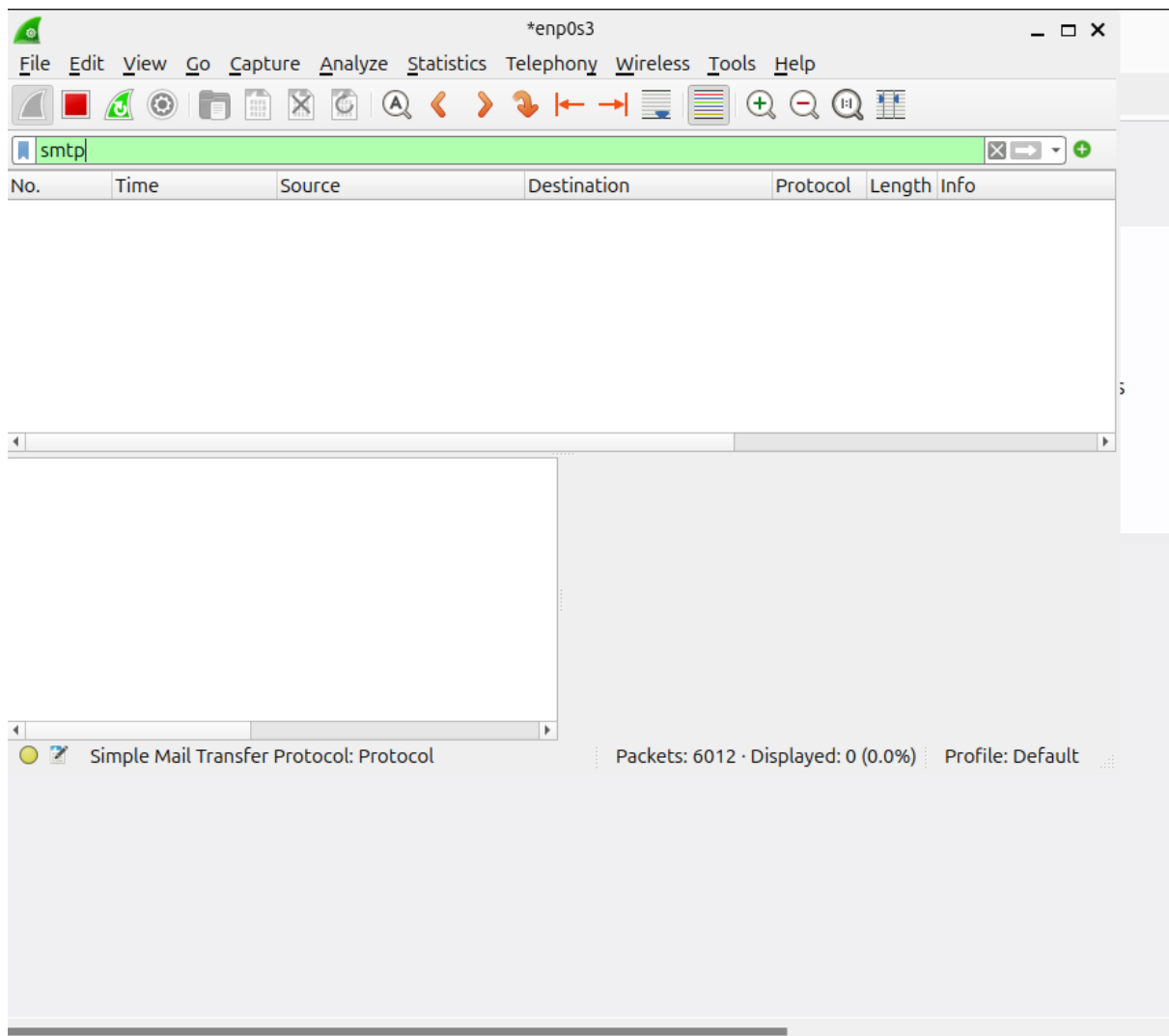


SARCINA 1: Interceptarea traficului HTTP necriptat





SARCINA 2: Analizarea traficului SMTP pentru vulnerabilități de e-mail



```
andreea@andreea-VirtualBox: ~  
The following NEW packages will be installed:  
  thunderbird  
1 upgraded, 1 newly installed, 0 to remove and 289 not upgraded.  
Need to get 25.5 kB of archives.  
After this operation, 72.7 kB of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 thunderbird amd64 2:  
1-0ubuntu3 [25.5 kB]  
Fetched 25.5 kB in 1s (35.0 kB/s)  
reconfiguring packages ...  
Selecting previously unselected package thunderbird.  
Reading database ... 149390 files and directories currently installed.)  
Preparing to unpack .../thunderbird_2%3a1snap1-0ubuntu3_amd64.deb ...  
> Installing the thunderbird snap  
=> Checking connectivity with the snap store  
=> Installing the thunderbird snap  
thunderbird 128.6.1esr-3 from Canonical✓ installed  
> Snap installation complete  
Unpacking thunderbird (2:1snap1-0ubuntu3) ...  
Setting up thunderbird (2:1snap1-0ubuntu3) ...  
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...  
Processing triggers for desktop-file-utils (0.27-2build1) ...  
andreea@andreea-VirtualBox:~$
```

Email address

rotaru.andreea.2001@gmail.com



Password

.....



☐ Remember password

✓ Configuration found in Mozilla ISP database.

Available configurations

☒ **IMAP**

Keep your folders and emails synced on your server

Incoming **IMAP** **SSL/TLS**

imap.gmail.com

Outgoing **SMTP** **SSL/TLS**

smtp.gmail.com

Username

rotaru.andreea.2001@gmail.com

☐ **POP3**

Keep your folders and emails on your computer

[Configure manually](#)

Cancel

Done

rotaru.andreea.2001@gmail.com Account Setup Thunderbird Privacy Notice

✓ Account successfully created

You can now use this account with Thunderbird.
You can improve the experience by connecting related services and configuring advanced account settings.

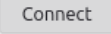
 **Rotaru Andreea** rotaru.andreea.2001@gmail.com **IMAP**

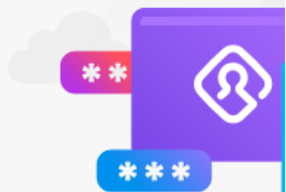
 Account settings  End-to-end encryption

 Add a signature  Download dictionaries

Connect your linked services

Thunderbird detected other services linked to your email account.

Address Books
Thunderbird found one address book linked to your email account. 
CARDDAV Address Book 



Not sure about your n
[Getting started](#) - [Support foru](#)

Write: test SMTP - Thunderbird

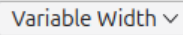
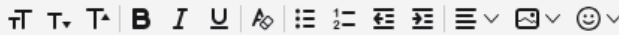
File Edit View Insert Format Options Tools Help

 Send  Encrypt  Spelling  Save  Contacts  Attach

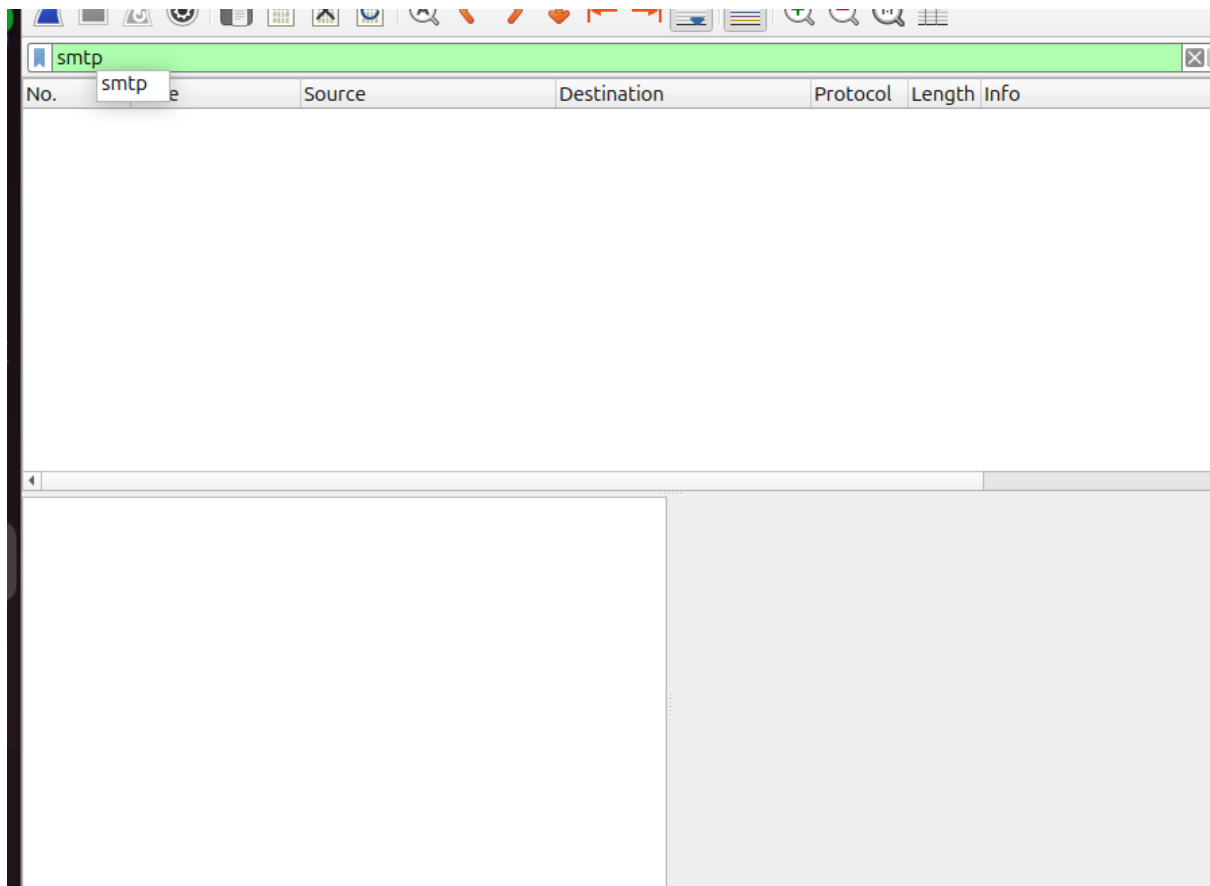
From Rotaru Andreea <rotaru.andreea.2001@gmail... rotaru.andreea.2001@gmail.com  | Cc Bcc

To rotaru.andreea.2001@gmail.com

Subject test SMTP

Body Text  Variable Width  

Aceasta este o captura SMTP pentru Wireshark



Gmail folosește **SSL/TLS pe portul 465**, ceea ce împiedică Wireshark să vadă conținutul mesajelor.

SMTP Server

Settings

Description:

Server Name:

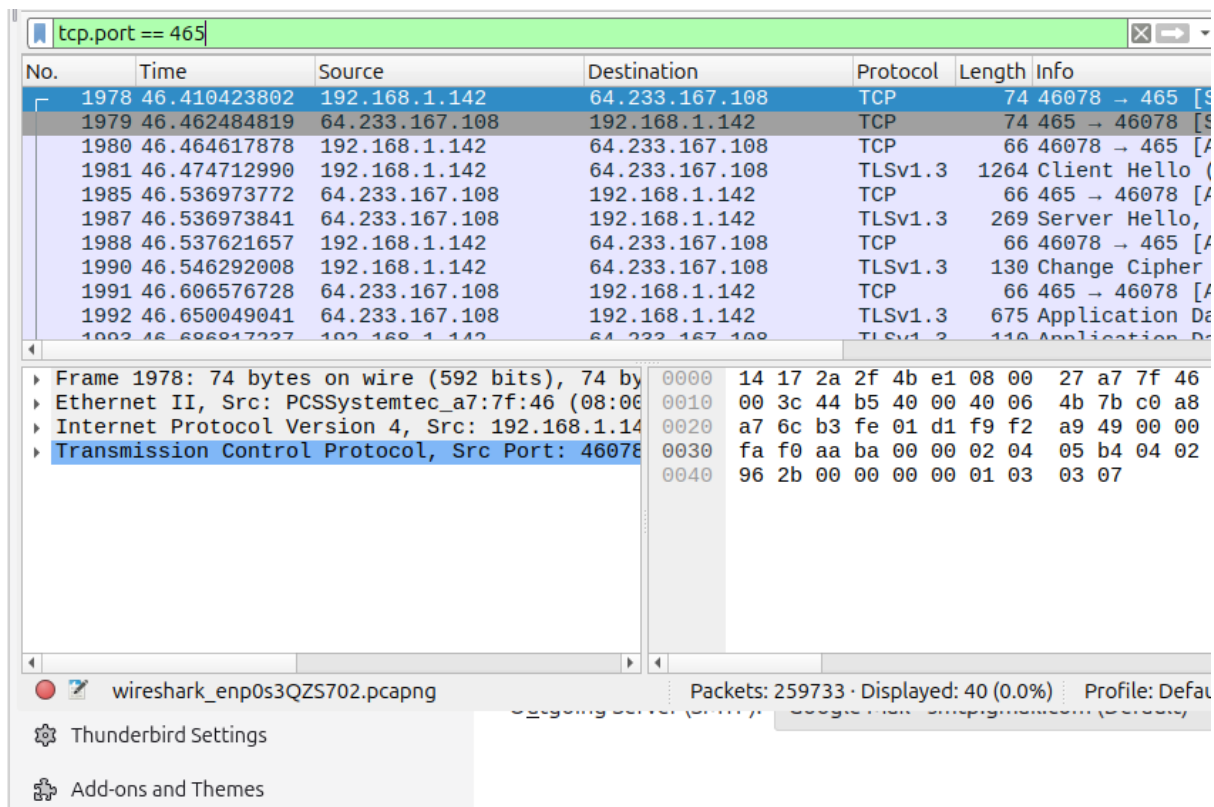
Port: Default: 465

Security and Authentication

Connection security:

Authentication method:

User Name:



Pachetele capturate în Wireshark arată conexiuni **TCP** și **TLSv1.3**, ceea ce indică faptul că emailurile sunt transmise în formă **criptată** și **nu pot fi citite**.

Sarcina 3: Identificarea activității neașteptate în propria rețea

```
andreea@andreea-VirtualBox:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:a7:7f:46 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.142/24 brd 192.168.1.255 scope global dynamic noprefixroute
        valid_lft 82338sec preferred_lft 82338sec
    inet6 fe80::a00:27ff:fea7:7f46/64 scope link
        valid_lft forever preferred_lft forever
andreea@andreea-VirtualBox:~$
```


ip.src == 192.168.1.142						
No.	Time	Source	Destination	Protocol	Length	Info
304	43.858974772	192.168.1.142	224.0.0.22	IGMPv3	54	Membership Report
1067	73.416879385	192.168.1.142	224.0.0.22	IGMPv3	54	Membership Report
1515	103.297305203	192.168.1.142	224.0.0.22	IGMPv3	54	Membership Report
1770	115.557996933	192.168.1.142	208.67.222.123	DNS	100	Standard query 0x
1772	115.615623067	192.168.1.142	185.125.190.97	TCP	74	35264 → 80 [SYN]
1775	115.660962881	192.168.1.142	185.125.190.97	TCP	66	35264 → 80 [ACK]
1776	115.662317290	192.168.1.142	185.125.190.97	HTTP	154	GET / HTTP/1.1
1780	115.705780345	192.168.1.142	185.125.190.97	TCP	66	35264 → 80 [ACK]
1781	115.707324902	192.168.1.142	185.125.190.97	TCP	66	35264 → 80 [FIN,
2416	163.176417768	192.168.1.142	224.0.0.22	IGMPv3	54	Membership Report

▶ Frame 212: 100 bytes on wire (800 bits), 100 B captured (800 bits) on interface 0 ▶ Ethernet II, Src: PCSSystemtec_a7:7f:46 (08:00:00:08:00:00), Dst: 01:00:5e:00:00:02 ▶ Internet Protocol Version 4, Src: 192.168.1.142, Dst: 208.67.222.123 ▶ User Datagram Protocol, Src Port: 33430, Dst Port: 53 ▶ Domain Name System (query)	0000 14 17 2a 2f 4b e1 08 00 27 a7 7f 46 08 00 00 0010 00 56 fd e9 00 00 40 11 0b b8 c0 a8 01 80 00 0020 de 7b 82 96 00 35 00 42 71 49 36 05 01 00 00 0030 00 00 00 00 00 01 12 63 6f 6e 6e 65 63 74 0040 69 74 79 2d 63 68 65 63 6b 06 75 62 75 6e 68 0050 03 63 6f 6d 00 00 1c 00 01 00 00 29 05 c0 0060 00 00 00 00
---	--

Identificarea activității neașteptate în propria rețea – Analiză Wireshark

În această sarcină, am utilizat Wireshark pentru a monitoriza și analiza traficul rețelei, având ca scop identificarea oricărei activități suspecte care ar putea indica o posibilă breșă de securitate. Am aplicat un filtru specific (ip.src == 192.168.1.142), unde 192.168.1.142 este adresa IP a dispozitivului utilizat, pentru a vizualiza doar traficul generat de acesta.

Analiza traficului capturat

În urma analizei pachetelor capturate, am identificat două tipuri principale de trafic: DNS (Domain Name System) și IGMP (Internet Group Management Protocol).

1. Pachet DNS către OpenDNS (208.67.222.123)

- Dispozitivul meu a trimis o solicitare DNS către serverul 208.67.222.123, care aparține OpenDNS.
- Acest comportament este unul obișnuit, deoarece DNS este utilizat pentru a rezolva numele de domenii în adrese IP.
- Totuși, merită verificat dacă acest server DNS a fost configurat intenționat, întrucât unele rețele folosesc Google DNS (8.8.8.8) sau serverele ISP-ului.

2. Pachet IGMP Membership Report către 224.0.0.22

- Protocolul IGMP este utilizat pentru gestionarea traficului multicast în rețea.
- Destinația 224.0.0.22 este o adresă multicast standard, folosită de routere pentru a menține grupurile multicast.

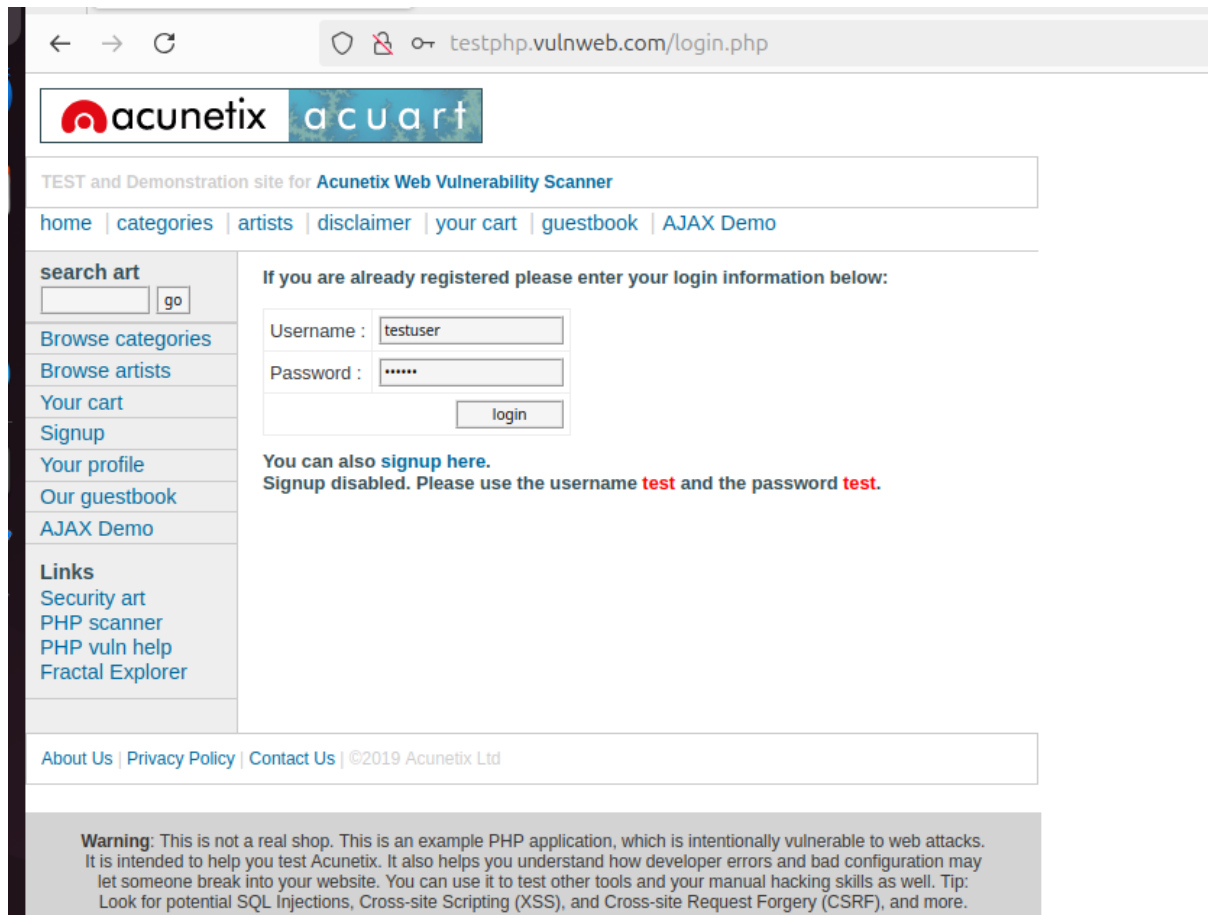
- Acest tip de trafic este întâlnit frecvent în rețelele care utilizează IPTV, Chromecast, servicii de streaming sau actualizări de sistem.

Concluzie

Analiza traficului capturat nu a indicat activități suspecte sau potențiale breșe de securitate. Toate pachetele detectate sunt normale și legate de funcționarea standard a rețelei.

- DNS-ul OpenDNS este un serviciu legitim, însă ar trebui verificat dacă a fost setat manual.
- Traficul IGMP este obișnuit în rețele care folosesc dispozitive multicast.
- Nu s-au detectat conexiuni către adrese IP necunoscute, porturi nesigure sau activități care ar putea indica un atac informatic.

Sarcina 4: Găsirea în traficul interceptat a unui login și a unei parole



The screenshot shows a web browser window with the address bar displaying `testphp.vulnweb.com/login.php`. The page header includes the Acunetix logo and the text "TEST and Demonstration site for Acunetix Web Vulnerability Scanner". Below the header, there is a navigation menu with links: home, categories, artists, disclaimer, your cart, guestbook, and AJAX Demo. On the left side, there is a sidebar with a search bar, a list of categories, and a list of links including Security art, PHP scanner, PHP vuln help, and Fractal Explorer. The main content area contains a login form with the following fields: Username (containing "testuser"), Password (containing "*****"), and a "login" button. Below the login form, there is a message: "You can also [signup here](#). Signup disabled. Please use the username **test** and the password **test**." At the bottom of the page, there is a footer with links: About Us, Privacy Policy, Contact Us, and a copyright notice: ©2019 Acunetix Ltd. A warning message is displayed at the very bottom: "Warning: This is not a real shop. This is an example PHP application, which is intentionally vulnerable to web attacks. It is intended to help you test Acunetix. It also helps you understand how developer errors and bad configuration may let someone break into your website. You can use it to test other tools and your manual hacking skills as well. Tip: Look for potential SQL Injections, Cross-site Scripting (XSS), and Cross-site Request Forgery (CSRF), and more."

*enp0s3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
2471	151.957232893	192.168.1.131	157.240.251.6	TCP	60	59987 → 443
2472	151.957233150	192.168.1.131	239.255.255.250	SSDP	179	M-SEARCH * H
2473	151.979020858	185.60.218.2	192.168.1.131	TLSv1.2	79	Application
2474	152.020047361	192.168.1.131	185.60.218.2	TCP	60	59979 → 443
2475	153.452028165	192.168.1.142	34.120.208.123	TLSv1.3	105	Application
2476	153.454681548	192.168.1.142	216.58.206.67	TCP	66	38344 → 80
2477	153.477330414	34.120.208.123	192.168.1.142	TLSv1.3	105	Application
2478	153.477357435	192.168.1.142	34.120.208.123	TCP	66	33176 → 443
2479	153.495474930	216.58.206.67	192.168.1.142	TCP	66	80 → 38344
2480	153.495502494	192.168.1.142	216.58.206.67	TCP	66	38344 → 80

Frame 1: 293 bytes on wire (2344 bits), 293 by
Ethernet II, Src: FiberhomeTel_2f:4b:e1 (14:17:00:01:83:01), Dst: 01:83:01:bb:f8:83
Internet Protocol Version 4, Src: 146.112.41.3, Dst: 192.168.1.131
Transmission Control Protocol, Src Port: 443, Dst Port: 80
Transport Layer Security

0000 40 1a 58 58 26 45 14 17 2a 2f 4b e1 08
0010 01 17 56 2f 40 00 3a 06 6c 13 92 70 29
0020 01 83 01 bb f8 83 b5 c2 52 e9 77 04 9f
0030 00 20 10 69 00 00 17 03 03 00 ea c9 60
0040 b5 c1 7d a0 0a 1f 1c 13 2d 2f f1 18 30
0050 b1 0e 93 1e 29 4a 64 ee b1 2b e9 d1 0b
0060 cb e3 b1 5d 26 47 2c ba 9a b5 95 41 52
0070 41 3f bf 8b a8 b6 bb 59 69 5f fc 7d 38
0080 12 9d 51 1b bc a7 02 7b 4d 6c b9 9d a5
0090 f2 17 ef cc 28 29 a7 a1 41 e3 1f 9d e7
00a0 26 6c 2a 27 d4 16 6f 50 7c 45 f1 9f eb
00b0 e0 c7 be 31 e5 8a 3e 11 33 18 1a 53 c8

wireshark_enp0s3M57Q02.pcapng
Packets: 2480 · Displayed: 2480 (100.0%) · Dropped: 0 (0.0%)
Profile: Default

Warning: This is not a real shop. This is an example PHP application, which is intentionally vulnerable to web attacks. It is intended to help you test Acunetix. It also helps you understand how developer errors and bad configuration may let someone break into your website. You can use it to test other tools and your manual hacking skills as well. Tip: Look for potential SQL Injections, Cross-site Scripting (XSS), and Cross-site Request Forgery (CSRF), and more.

